

Impact of Regulatory Reforms on Formula 1 Performance: A Data-Driven Analysis

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1 Introduction

Formula 1 is the technological pinnacle of motorsport, where performance depends on a balance between engineering expertise, financial resources, and regulatory interpretation. During the hybrid era (2014–2021), teams with greater technical and economic capabilities, such as Mercedes, have managed to build long-term competitive dominance.

In recent years, the FIA has reshaped this balance through significant regulatory reforms. The reintroduction of ground effect in 2022 has modified the cars' aerodynamics, reducing wake turbulence and favoring overtaking. At the same time, safety regulations have been strengthened, particularly with the introduction of the Halo in 2018, along with more strict crash tests and structural standards.

At the same time, the budget cap, introduced in 2021, has limited teams' annual spending, reducing economic disparities and forcing teams to prioritize efficiency over pure investment. Overall, these changes have radically altered the way performance is achieved in Formula 1, leading to the central research question: "How have ground effects regulations and budget constraints affected the performance of Formula 1 cars?"

To answer this question through a data-driven methodology, the project integrates multiple data management techniques for data extraction and analysis, such as scraping, APIs, OpenAI and SQL. The system's architecture is based on technical pillars, each designed to address specific challenges in data collection, processing, and analysis.

2 Data Acquisition - Source 1: Qualifying Results (Jolpica API)

The first pillar consists of data acquisition via the Jolpica public API, which provides structured access to the entire Formula 1 qualifying results history.

Qualifying follows a three-session knockout format: Q1 eliminates the five slowest drivers from 20 competitors, Q2 eliminates another five from the remaining 15, and Q3 sees the 10 fastest drivers compete for pole position. This generates a hierarchical data structure where those eliminated in Q1 have only that time, those who reach Q2 have two times, and only the top 10 have complete times across all three sessions.

This technique implements robust pagination patterns to manage large data volumes, in our case over 3,800 qualifying results, with rate-limiting logic to comply with API constraints and strategies to handle structural missing data, such as Q2 and Q3 times for drivers eliminated in previous rounds.

The choice to focus on the Q1, Q2 and Q3 data is strategic: it represents the maximum performance and highest potential of the cars, minimizing race variables such as strategy, mechanical reliability, limited amount of fuel for less weight, and accidents. So, finally we got the qualification data with API technique.

Results: Qualifications Dataframe

For every year we have:

- 2017: 20 races, 398 results
- 2018: 21 races, 420 results
- 2019: 21 races, 418 results
- 2020: 17 races, 340 results
- 2021: 22 races, 439 results
- 2022: 22 races, 440 results
- 2023: 22 races, 440 results
- 2024: 24 races, 479 results
- 2025: 24 races, 479 results

Total: 3853 qualifying results

3 Entity Resolution - Team Name Mapping

The second pillar addresses a crucial challenge in sports data management: entity resolution for team names. Formula 1 is characterized by frequent name changes for commercial reasons, for instance:

- Racing Point becomes Aston Martin
- Renault becomes Alpine
- Toro Rosso becomes Alpha Tauri and then RB

This makes longitudinal analysis impossible without systematic mapping. The project implements an entity resolution dictionary that tracks the organizational and competitive continuity of teams across rebrandings, ensuring that analyses such as “Aston Martin performance 2017–2025” correctly include Force India and Racing Point as predecessors of the same competitive entity.

Results

We obtain a new column in the dataset called `Team_Entity` that contains the canonical name of the competitive entity, alongside the original `Team` column from the API.

Table 1: Sample Data: Team Entity Mapping (2018 Australian GP)

Year	Pos	Driver	Team	Q1	Q2	Q3	Team Entity
2018	1	HAM	Mercedes	1:22.824	1:22.051	1:21.164	Mercedes
2018	2	RAI	Ferrari	1:23.096	1:22.507	1:21.828	Ferrari
2018	3	VET	Ferrari	1:23.348	1:21.944	1:21.838	Ferrari
2018	4	VER	Red Bull	1:23.483	1:22.416	1:21.879	Red Bull
2018	5	RIC	Red Bull	1:23.494	1:22.897	1:22.152	Red Bull
2018	6	MAG	Haas F1 Team	1:23.909	1:23.300	1:23.187	Haas F1 Team
2018	7	GRO	Haas F1 Team	1:23.671	1:23.468	1:23.339	Haas F1 Team
2018	8	HUL	Renault	1:23.782	1:23.544	1:23.532	Alpine F1 Team
2018	9	SAI	Renault	1:23.529	1:23.061	1:23.577	Alpine F1 Team
2018	10	BOT	Mercedes	1:23.686	1:22.089	None	Mercedes
2018	11	ALO	McLaren	1:23.597	1:23.692	None	McLaren
2018	12	VAN	McLaren	1:24.073	1:23.853	None	McLaren
2018	13	PER	Force India	1:24.344	1:24.005	None	Aston Martin
2018	14	STR	Williams	1:24.464	1:24.230	None	Williams
2018	15	OCO	Force India	1:24.503	1:24.786	None	Aston Martin
2018	16	HAR	Toro Rosso	1:24.532	None	None	RB F1 Team
2018	17	ERI	Sauber	1:24.556	None	None	Sauber
2018	18	LEC	Sauber	1:24.636	None	None	Sauber
2018	19	SIR	Williams	1:24.922	None	None	Williams
2018	20	GAS	Toro Rosso	1:25.295	None	None	RB F1 Team

4 LLM Regulations Extraction

The third pillar leverages an LLM-based approach to extract historical regulatory data. Using `gpt-5-mini-2025-08-07`, the system extracted and structured the entire FIA regulatory evolution from 2017 to 2025, categorizing each regulatory change by type:

- aerodynamic
 - financial
 - safety
 - engine
- and impact:
- revolutionary
 - major
 - minor

This data retrieval technique transforms complex, unstructured textual documents into parsable datasets. The LLM-based approach overcomes the limitations of web scraping for complex content such as sporting regulations, where deep semantic understanding is required to extract relevant information. To receive data in a structured way in the prompt was added to return specifically JSON array type:

```

1 prompt = f''' For {year} F1 season, list main regulation changes.
2 Return JSON array: [{"Year": {year}, "Type": "Aero/Financial/Safety/Engine",
3 "Description": "...", "Impact": "Major/Minor/Revolutionary",

```

```

4 "Source": "FIA {year}"]}]".
5 Only JSON.'''

```

Listing 1: Sytem prompt for OpenAI API call

Results

Via `gpt-5-mini-2025-08-07` you obtain a structured dataset of 46 regulations covering all FIA reforms from 2017 to 2025.

Table 2: LLM Regulations Extraction Results (2017)

Year	Type	Description	Impact	Source	Era	Extraction Method
2017	Aero	Comprehensive aerodynamic overhaul for 2017: c...	Revolutionary	FIA 2017	Pre-2022	OpenAI API (gpt-5-mini-2025-08-07)
2017	Aero	Tyre and wheel regulation changes: significant...	Major	FIA 2017	Pre-2022	OpenAI API (gpt-5-mini-2025-08-07)
2017	Aero	Car dimensional and mass adjustments: overall ...	Major	FIA 2017	Pre-2022	OpenAI API (gpt-5-mini-2025-08-07)
2017	Engine	Power unit rules largely carried over from pri...	Minor	FIA 2017	Pre-2022	OpenAI API (gpt-5-mini-2025-08-07)
2017	Safety	Incremental safety rule updates: ongoing tight...	Minor	FIA 2017	Pre-2022	OpenAI API (gpt-5-mini-2025-08-07)
2017	Financial	No new cost-cap or major financial redistribut...	Minor	FIA 2017	Pre-2022	OpenAI API (gpt-5-mini-2025-08-07)

At the end of LLM search the results are saved in a CSV (`f1_regulations_openai.csv`) file to keep the records even if the kernel is rebooted and the results are lost since they were saved on the RAM and not on physical memory.

```

1 # Save the dataframe to avoid re-running the API call
2 df_regulations.to_csv('f1_regulations_openai.csv', index=False)

```

Listing 2: Exporting Regulations to CSV

5 ATR Allocations (Jolpica API)

The fourth pillar dynamically calculates ATR (Aerodynamic Testing Restrictions) allocations derived from API data from the constructors' standings. Since 2022, the FIA has introduced a scaled testing restriction system based on two fundamental aerodynamic development tools:

- **Wind Tunnel:** A facility that artificially recreates controlled airflow up to 112mph to test scale models of single-seaters and measure aerodynamic forces, drag, and downforce.
- **CFD (Computational Fluid Dynamics):** Simulations which virtually replicate the same phenomena through complex mathematical calculations.

The mechanism is highly asymmetric: the world champion team receives only 70% of the base wind tunnel and CFD simulation time, while the last-placed team receives 115%, with linear gradations for intermediate positions.

Table 3: 2022 ATR Allocations Sample

Note: ATR (Aerodynamic Testing Restrictions) based on previous year's championship position.

Year	Team	Prev. Pos	ATR %	WT Runs/Period	WT Annual	CFD Items/Period	CFD Annual	Source
2022	Mercedes	1	70.0	224	1344	1400	8400	Jolpica + FIA ATR
2022	Red Bull	2	75.0	240	1440	1500	9000	Jolpica + FIA ATR
2022	Ferrari	3	80.0	256	1536	1600	9600	Jolpica + FIA ATR
2022	McLaren	4	85.0	272	1632	1700	10200	Jolpica + FIA ATR
2022	Alpine F1 Team	5	90.0	288	1728	1800	10800	Jolpica + FIA ATR

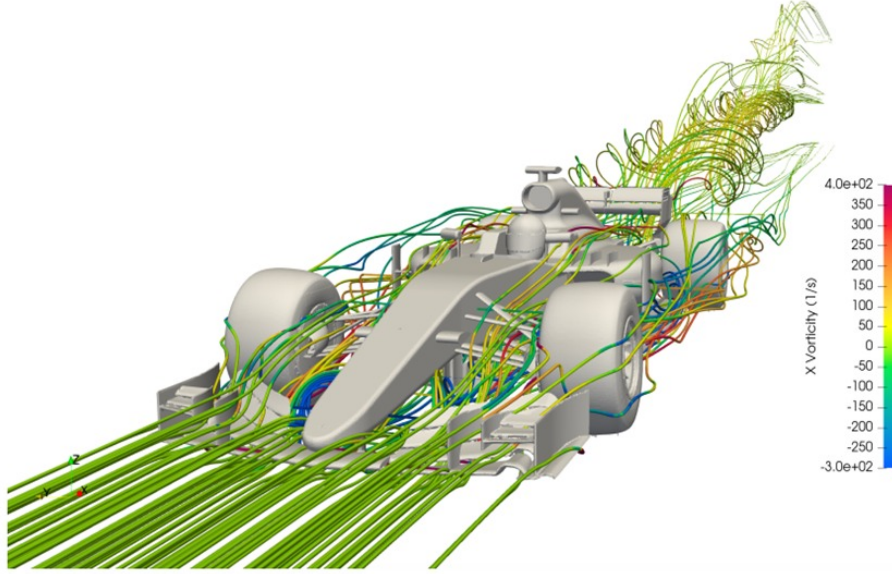


Figure 1: CFD Example

6 Data Preparation

The Data Preparation section takes the raw data acquired from the API and transforms it into parsable formats, solving the fundamental problem of qualifying times arriving as text strings (“1:23.456”) that are unusable for mathematical calculations and statistical comparisons.

The process involves four sequential transformations:

1. **Time Parsing:** A time parsing function is implemented that converts each M:SS.mmm (minutes:seconds.milliseconds) format string into a numeric value of decimal seconds: for example, “1:23.456” becomes 83.456 seconds, enabling subtraction, averaging, and sorting. This conversion also handles null values (eliminated drivers), returning None instead of generating errors.
2. **Pole Position Identification:** The system identifies the pole position time for each race by calculating the minimum of all converted times grouped by year and race round, creating a PoleTime column that represents the absolute performance benchmark for that specific event.
3. **Gap Calculation:** The third step calculates the absolute gap to the pole position by subtracting the pole time from each individual time: a driver with a time of 83.456 seconds in a race where the pole time is 82.000 seconds will have a gap of 1.456 seconds. This normalized metric allows for cross-circuit comparisons, eliminating the intrinsic differences between fast (Monza) and slow (Monaco) tracks.
4. **Era Labeling:** A categorical “Era” column is added, labeling each record as Pre-2022 or Post-2022, directly enabling comparative analyses between the two regulatory eras through simple group-by operations.

These transformations convert textual and hierarchical data into a dataset ready for statistical aggregations.

Results

After the transformations you get the `df_qualifying` dataset enriched with 4 new analytical columns:

```
article [utf8]inputenc geometry booktabs graphicx float pdfscape
a4paper, margin=1cm
```

Table 4: Data Preparation Results Sample (Monza) - Pre vs Post 2022 Era
Note: Comparison between 2021 (Pre-Regulation) and 2025 (Post-Regulation) qualifying times.

Year	Rnd	Grand Prix	Circuit	Pos	Driver	Team	Q1	Q2	Q3	BestTime	Best_Sec	PoleTime	Gap	Era
2021	14	Italian GP	Monza	1	BOT	Mercedes	1:20.685	1:20.032	1:19.555	1:19.555	79.555	79.555	0.000	Pre-2022
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2021	14	Italian GP	Monza	9	PER	Red Bull	1:21.308	1:20.882	1:20.611	1:20.611	80.611	79.555	1.056	Pre-2022
2021	14	Italian GP	Monza	10	GIO	Alfa Romeo	1:21.197	1:20.726	1:20.808	1:20.808	80.808	79.555	1.253	Pre-2022
2025	16	Italian GP	Monza	1	VER	Red Bull	1:19.455	1:19.140	1:18.792	1:18.792	78.792	78.792	0.000	Post-2022
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2025	16	Italian GP	Monza	9	ALO	Aston Martin	1:19.658	1:19.362	1:19.424	1:19.424	79.424	78.792	0.632	Post-2022
2025	16	Italian GP	Monza	10	TSU	Red Bull	1:19.619	1:19.433	1:19.519	1:19.519	79.519	78.792	0.727	Post-2022

7 Data Integration

The Data Integration section represents the point where all the separately collected datasets are unified into a single structure, creating an interconnected system where each qualifying result carries its own regulatory context and restrictions.

The process begins by aggregating the 46 individual regulations into a summary by year that lists how many reforms were introduced in total, how many were classified as Major or Revolutionary, and whether that year saw revolutionary changes. For example, 2022 is labeled with `RegulationChanges=8`, `MajorChanges=6`, `HasRevolutionary=Yes` because it saw the introduction of the ground effect, simplified wings to get cleaner slipstream to promote closer racing.

This summary is then merged with the qualifying dataset on Year, automatically propagating the regulatory information to all 3,853 qualifying results: every record of Verstappen’s pole position in Bahrain 2024 now “knows” that season had four regulations, including two major ones.

At the same time, the ATR allocations are merged into the dataset by Year and Team, linking each qualifying result to that team’s specific development restrictions for that season. For instance if we consider Monza 2022 after the column `HasRevolutionary` it will be displayed `ATR_Percentage` and all the other metrics about the aerodynamic R&D constraints.

Table 5: Integrated Dataset (Monza 2022) - Full Data Integration
Note: some columns have been compacted for readability.

Year	Round	GrandPrix	Circuit	Pos	Driver	Team	BestTime	Gap	Era	Regs (M/R)	ATR%	WT/CFD
2022	16	Italian GP	Monza	1	LEC	Ferrari	1:20.161	0.000	Post-22	8 (6/Yes)	80.0	1536/9600
2022	16	Italian GP	Monza	2	VER	Red Bull	1:20.306	0.145	Post-22	8 (6/Yes)	75.0	1440/9000
2022	16	Italian GP	Monza	3	SAI	Ferrari	1:20.429	0.268	Post-22	8 (6/Yes)	80.0	1536/9600
2022	16	Italian GP	Monza	4	PER	Red Bull	1:21.206	1.045	Post-22	8 (6/Yes)	75.0	1440/9000
2022	16	Italian GP	Monza	5	HAM	Mercedes	1:21.524	1.363	Post-22	8 (6/Yes)	70.0	1344/8400
2022	16	Italian GP	Monza	6	RUS	Mercedes	1:21.542	1.381	Post-22	8 (6/Yes)	70.0	1344/8400
2022	16	Italian GP	Monza	7	NOR	McLaren	1:21.584	1.423	Post-22	8 (6/Yes)	85.0	1632/10200
2022	16	Italian GP	Monza	8	RIC	McLaren	1:21.925	1.764	Post-22	8 (6/Yes)	85.0	1632/10200
2022	16	Italian GP	Monza	9	GAS	AlphaTauri	1:22.648	2.487	Post-22	8 (6/Yes)	95.0	1824/11400
2022	16	Italian GP	Monza	10	ALO	Alpine	1:21.861	1.700	Post-22	8 (6/Yes)	90.0	1728/10800



Figure 2: 2021 VS 2022 example

8 Data Quality Assessment

The pipeline includes the handling of structural nulls/missing values that are not related to data quality issues but rather inherent to the data structure. In F1 qualifying, only 15 drivers advance to Q2, and only 10 reach Q3. This means that approximately 25% of drivers have null Q2 times and 50% have null Q3 times by design, not due to data collection failures.

The function accepts an optional `exclude_cols` parameter that allows specific columns to be excluded from the overall completeness calculation while still reporting their individual completeness rates. For the qualifying dataset, Q2 and Q3 are excluded from the overall metric, since it is enough to verify if there is a record in Q1 to check if the event took apart.

The pipeline produces three outputs:

1. Per-column completeness percentages (showing which fields have missing data)
2. Overall completeness metrics (excluding structural nulls)
3. A summary DataFrame comparing the three primary datasets (Qualifying, Regulations, ATR Allocations)

Table 6: Data Quality Assessment Summary

Note: Completeness metrics exclude structural nulls (e.g., Q2/Q3 for eliminated drivers).

Dataset	Records	Columns	Completeness (%)
Qualifying	3853	24	99.80%
Regulations	46	7	100.00%
ATR Allocations	40	9	100.00%

9 Analysis - Field Spread by Era

The Field Spread analysis quantifies competitive balance in Formula 1 by measuring the time gap between the pole position (P1) and the 10th-fastest qualifier (P10) for each event. This

metric, named “field spread,” indicates how close the top teams are in performance: smaller gaps suggest tighter competition, while larger gaps indicate dominance by a few teams.

The analysis methodology involves three steps:

1. The dataset is filtered to include only the top 10 qualifiers per race (positions 1–10).
2. For each race, the system calculates the maximum gap to pole position among these top 10 drivers, which represents the P1–P10 spread.
3. These spreads are aggregated by year and by era (Pre-2022 vs. Post-2022) to identify trends.

The visualization component generates two complementary charts:

- A yearly bar chart showing the evolution of field spread from 2017–2025, with color coding to differentiate pre-ground effect and post-regulation eras and a vertical line marking the 2022 regulatory boundary.
- An era comparison chart displaying the average spread for each era with numerical annotations.

The Pre-2022 era shows an average P1-P10 gap of 2.72 seconds, while the Post-2022 era (ground effect + budget cap) reduces this to 2.15 seconds, a 22% gap reduction. The trend accelerates over time: 2025 shows the tightest field in the analysis period with only 1.11 seconds average gap, suggesting that the budget cap’s equalizing effect increases as teams converge toward the spending limit.

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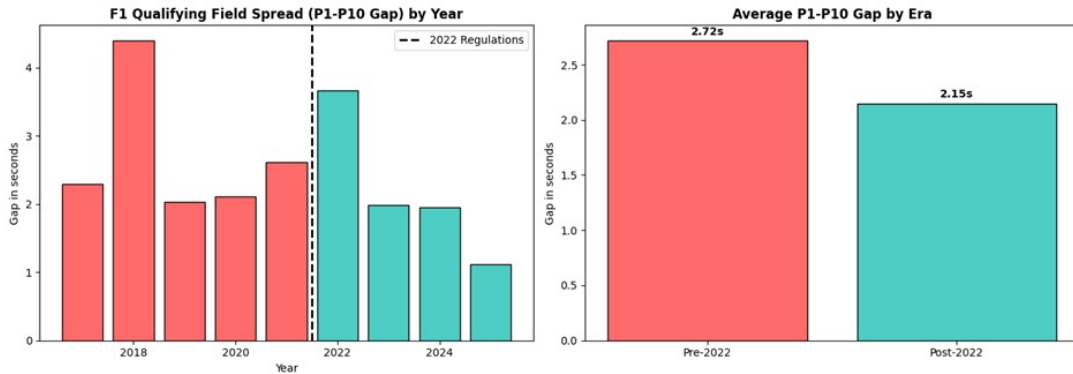


Figure 3: Field Spread Analysis: P1-P10 Gap Evolution by Era

This quantitative evidence supports the hypothesis that the 2022 regulatory package successfully achieved its stated goal of improving competitive parity in Formula 1.

10 Storage & Export

After evaluating the ground effect new regulation, and after all the integration done to make the data complete as much as possible for the research intent, it is a good practice to save the dataframes into CSV to avoid running multiple times the notebook and to avoid the call over and over OpenAI API consuming tokens.

The following files are exported:

- `f1_qualifying_complete.csv`
- `f1_regulations_changes.csv`
- `f1_atr_allocations.csv`
- `f1_field_spread.csv`
- `f1_data_quality_report.csv`

11 Source 4: WDC/WCC Championship Winners (Jolpica API)

The fourth data source captures the World Drivers Championship (WDC) winner, the individual driver with the most points, -and the World Constructors Championship (WCC) winner—the team with the highest combined points from both drivers. Technically there could be more than 2 drivers for a team; for instance, Ferrari during Jeddah GP in 2024 when C. Sainz had to go through a surgery, Ollie Bearman took his place for the event and we have the sum of the points of 3 drivers not just 2.

The implementation uses two Jolpica API endpoints:

- `/driverstandings/1.json` retrieves the championship leader (position 1 in final standings) for a given year, returning the driver's name and their team.
- `/constructorstandings/1.json` retrieves the constructor championship winner for that season.

This data serves two critical analytical purposes:

1. **Regulatory Impact Assessment:** By correlating championship outcomes with regulatory eras, the analysis can determine whether the 2022 regulations disrupted existing competitive hierarchies (they did since Mercedes lost WCC dominance to Red Bull after 8 years of absolute outstanding performance).
2. **Entity Resolution Verification:** Confirming that team names in championship data match the standardized entities used in qualifying data ensures join accuracy across datasets.

An interesting pattern emerges in the data: from 2017–2021, Mercedes won every WCC despite losing the 2021 WDC to Max Verstappen. Post-2022, Red Bull dominated both championships (2022–2023), but by 2024, McLaren won the WCC thanks also to a more balanced performance of the drivers inside the team, while Verstappen retained the WDC, indicating that the budget cap successfully prevented sustained dual-championship monopolies.

12 SQL Storage (SQLite)

The choice of SQL for the DB was made to let the most of the users work on it. It is also fast and for the data structure we have it doesn't require much work around. It is also due to the nature of data itself since for every event, the GP qualify we have 20 drivers competing each other and then we extract a poleman with all the information needed. For a wider overview we can say every year has a certain amount of races which have the qualify session with the poleman and lap time; even though we don't just collect the poleman lap time but the top 10 driver's lap time to get a better comparison and to figure out if the competition gets tighter or not.

To ensure we are using a fresh DB the first step is deleting any existing database file, just to keep a clean state and not risk working with an old database from a previous run. Then the

tables are created using `pandas.to_sql()`, which automatically creates the tables and infers the schema/types, so there's no need to manually write `CREATE TABLE` statements. This keeps the code shorter and lets the user build the DB + tables faster.

The SQL Storage component basically takes the project flat files/DataFrames and converts everything into a relational database, so we can run real SQL queries and not just do everything in pandas. SQLite was selected because it's zero-config, portable (single db file), and still gives full SQL support.

The database is made by 5 tables:

1. **seasons:** 9 records, columns: year, wdc_driver, wdc_team, wcc_team. This is the “timeline” table for the whole DB.
2. **regulations:** 46 records, columns: year, type (Aero/Financial/Engine/Safety), description, impact (Revolutionary/Major/Minor), source, era. The era tag (pre/post 2022) makes it easy to run queries like “show all revolutionary regs in the post-2022 era”.
3. **events:** 233 records, race-level data: year, round, GP name, circuit + pole info (pole_time, pole_driver, pole_team), taken from qualifying results where Position = 1.
4. **teams:** 90 records, columns: year, name, is_wcc (1 if WCC winner), atr_percentage, wt_hours_annual. It's 90 because we have 9 seasons $\times$ 10 teams (9 $\times$ 10).
5. **drivers:** 198 records, columns: year, driver code, full name, team, is_wdc, races_count, first_round, last_round. This is useful to analyze driver participation patterns and link them to championship outcomes.
6. **circuit performance:** 9 records .

To test the DB, two demo queries have been used:

1. The first one retrieved the WDC/WCC winners query (2017–2025) using the seasons table.
2. The second one gives back the regulations summary query, grouped by year, using `COUNT(*)` + `GROUP_CONCAT` to list the regulation types.

```
1 SELECT year, wdc_driver, wdc_team, wcc_team
2 FROM seasons
3 ORDER BY year
```

```
1 SELECT
2     year,
3     COUNT(*) as total,
4     GROUP_CONCAT(DISTINCT type) as types,
5     SUM(CASE WHEN impact = 'Major' THEN 1 ELSE 0 END) as major_count,
6     SUM(CASE WHEN impact = 'Revolutionary' THEN 1 ELSE 0 END) as
    revolutionary_count
7 FROM regulations
8 GROUP BY year
```

Table 7: SQL Query Results: WDC & WCC Winners (2017-2025)
Note: Years with split championships (Driver vs Constructor) are highlighted in bold.

Year	WDC Driver	Driver Team	WCC Team	Era
2017	Lewis Hamilton	Mercedes	Mercedes	Pre-2022
2018	Lewis Hamilton	Mercedes	Mercedes	Pre-2022
2019	Lewis Hamilton	Mercedes	Mercedes	Pre-2022
2020	Lewis Hamilton	Mercedes	Mercedes	Pre-2022
2021	Max Verstappen	Red Bull	Mercedes	Pre-2022
2022	Max Verstappen	Red Bull	Red Bull	Post-2022
2023	Max Verstappen	Red Bull	Red Bull	Post-2022
2024	Max Verstappen	Red Bull	McLaren	Post-2022
2025	Lando Norris	McLaren	McLaren	Post-2022

Table 8: SQL Query Results: Regulations Summary & Impact Analysis
Note: "Rev" indicates Revolutionary changes. The 2022 spike marks the Ground Effect introduction.

Year	Total Regs	Major	Rev	Key Areas	Period
2017	6	2	1	Aero, Engine, Safety, Financial	Pre-2022
2018	5	1	0	Safety, Engine, Aero, Financial	Pre-2022
2019	4	2	0	Aero, Safety, Engine, Financial	Pre-2022
2020	4	0	0	Aero, Financial, Safety, Engine	Pre-2022
2021	4	3	0	Financial, Engine, Safety, Aero	Pre-2022
2022	8	5	1	Aero, Financial, Safety, Engine	Post-2022
2023	7	4	0	Aero, Financial, Safety, Engine	Post-2022
2024	4	2	0	Financial, Engine, Aero, Safety	Post-2022
2025	4	0	0	Aero, Financial, Safety, Engine	Post-2022

13 Conclusions

The conclusion of this pipeline which involves calls to LLM API for advanced data retrieving process, Jolpica API data retriving is: For what concerns how the budget-cap and various changes into the regulation affected the competitiveness between teams and narrowed the gap between the pole man and the 10th qualified, we came up to that conclusion also the gap in qualifying got narrow and narrow expecially at the end of the tecnical cycle, we sure have some outlier such as the 2018, but the reason behind was due to Williams Racing Team was having severe financial problems in fact they used to repair the car's floor with american duct tape. For the new era, post-2021, we can appreciate how the bar chart in figure 3 show a trend where the gaps reduces year after year so this is a point in favor to our thesis: the Budget cap and ATR for RD makes the grid and competition thighter.

14 EXTRA:

Since it wasn't required by the aim of the project we decided to show a solution to the question search: looking at the comparison in some specific circuits where minor changes have been made and no layout changes have been made. The results show smaller improvement in high speed tracks, and worse performance in slow tracks with sharp corners, this is also caused by the base concept of the floor effect: High speed is needed to develop a great ammount of downforce, not only that cars got havier by 40/45kgs which means in slow section is more tricky to find the best mechanical grip and this is a downside in "slow tracks" as Hungaroring or in tracks with sharp corners such as Redbull Ring.

The query used:

```
1 SELECT Circuit ,
2     ROUND(Pre_Best , 3) as Pre_2021 , Pre_Year ,
3     ROUND(Post_Best , 3) as Post_2022 , Post_Year ,
4     ROUND(Delta , 3) as Delta , Era_Diff
5 FROM circuit_performance ORDER BY Delta
```

Results:

Table 9: Circuit Performance Analysis: Pre-2022 vs Post-2022 Era

Note: Comparison of best qualifying lap times (in seconds). Negative delta indicates the new era is faster.

Circuit	Pre Time (s)	Year	Post Time (s)	Year	Delta (s)	% Diff	Verdict
Spa	101.252	2020	100.562	2025	-0.690	-0.68%	FASTER
Monaco	70.166	2019	69.954	2025	-0.212	-0.30%	FASTER
Monza	78.887	2020	78.792	2025	-0.095	-0.12%	FASTER
Suzuka	87.064	2019	86.983	2025	-0.081	-0.09%	FASTER
COTA	92.029	2019	92.330	2024	+0.301	+0.33%	SLOWER
Mexico City	74.758	2019	75.586	2025	+0.828	+1.11%	SLOWER
Red Bull Ring	62.939	2020	63.971	2025	+1.032	+1.64%	SLOWER
Hungaroring	73.447	2020	75.227	2024	+1.780	+2.42%	SLOWER
Bahrain	87.264	2020	89.179	2024	+1.915	+2.19%	SLOWER

Table 10: Performance Summary: Speed Change Overview

Note: "Avg Delta" represents the mean time difference across all analyzed circuits.

Category	Circuits	Avg Delta (s)	Trend
FASTER	4	-0.269	Improved high-speed efficiency
SLOWER	5	+1.171	Impact of weight & stiffness